Uganda Christian University

Bachelor of Science in Data Science and Analytics

Seven-Day Product Mining and Analysis of Jumia Nigeria

with a focused Kenya–Nigeria comparison

Course: DSC2107 Data Mining and Wrangling
Class: Year 2, Semester 1
Group: Group 2
Collection period: 13–19 October 2026
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Group members

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Declaration. This report summarises genuine public product observations collected during the stated dates. Raw files were retained unchanged, missing values were preserved, and no observations were fabricated or backdated.

1 Executive summary

This project examined how product prices, discounts, ratings, popularity and availability changed on Jumia Nigeria over seven consecutive days. Public catalogue pages were collected for *Phones & Tablets*, *Computing* and *Electronics*. The final analytical dataset contains 4,313 product-date observations, 799 unique products and seven genuine collection dates. A normalized SQLite database supported repeatable analysis and protected the raw evidence from accidental editing.

The main result is that prices were more likely to fall than rise. Among 715 products with comparable first and last observations, 252 (35.2%) decreased, 48 (6.7%) increased and 415 (58.0%) remained unchanged. In total, 314 products changed price at least once. Price had only a very weak positive association with rating ($\rho = 0.110$), while review count had a weak negative association with rating ($\rho = -0.215$). These relationships are descriptive, not causal.

A focused comparison with Group 1 used USD product medians for Jumia Kenya and Nigeria. The 38 closely matched listings had a median Nigeria-to-Kenya price ratio of 0.976, meaning the typical matched Nigerian listing was about 2.4% cheaper. However, country-level product mixes differed considerably; therefore, matched listings are more reliable than comparing overall averages.

2 Objectives and analytical questions

The project aimed to build a responsible collection-to-analysis workflow and answer the assignment questions at an appropriate undergraduate level:

- describe price distributions and identify category outliers;
- examine price–rating and review-count–rating relationships;
- track seven-day changes in prices and displayed discounts;
- identify dominant brands and the share of discounted products;
- investigate whether repeated visibility is related to price stability; and
- compare a focused subset of Jumia Nigeria and Jumia Kenya products in USD.

3 Scope

The Nigerian study covered server-rendered catalogue listings in three categories from 13 to 19 October 2026. A product-date row is an observation, not a sale. The study therefore describes advertised listings visible to the collector rather than Jumia's full inventory, transaction volume or consumer demand.

4 Methodology

4.1 Reconnaissance and responsible collection

Before scraping, the team checked the live page structure and `robots.txt`, confirmed that product cards were available in HTML, and recorded suitable selectors. Python `requests` downloaded pages sequentially and Beautiful Soup parsed the returned HTML. The collector used a descriptive user agent with a runtime contact address, a three-second delay, checkpoints, bounded retries and immediate stopping rules for CAPTCHA, HTTP 403, persistent HTTP 429 or other access-denial signals. It did not log in, bypass controls or collect personal information.

4.2 Collection pipeline

Each run followed the same reproducible sequence:

1. collect public product cards and save one accepted raw CSV for the true date;
2. validate required fields, timestamps, uniqueness and the minimum row target;
3. calculate and record a SHA-256 checksum for the accepted raw file;
4. clean into a separate file while leaving the raw file unchanged;
5. rebuild the SQLite database from all accepted cleaned daily files; and
6. execute the notebook and interpret tables and charts.

4.3 Cleaning and transformation

Product identifiers and URLs were normalized, text fields were trimmed, and numeric prices, ratings, review counts and discounts were parsed. Missing values were retained rather than replaced by zero. Duplicate product-date records were rejected. The final combined analytical CSV stores prices in USD to support cross-country comparison, while the original Nigerian raw evidence remains unchanged in NGN. The same conversion treatment was applied consistently before comparative analysis; percentage discounts are currency-independent.

4.4 Database design and analysis

The SQLite database separates categories, products, product–category relationships and daily observations. This avoids repeating stable product information and enforces one observation per product per date. The notebook used pandas for tables, Matplotlib/Seaborn for visualisation and Spearman correlation for monotonic relationships. Medians and log-scale boxplots were preferred for prices because a small number of expensive products created strong right-skew and outliers.

5 Quality controls

Eight unit tests checked numeric parsing, discount calculations, URL normalization, page construction, checkpoint loading, HTML parsing and duplicate database observations. All eight passed. SQLite reported integrity `ok`, zero foreign-key violations and zero duplicate product-date groups. The executed notebook completed 15 of 15 code cells with no saved errors and eight figures.

6 Data overview

Table 1: Accepted observations by collection date

Date	13 Oct	14 Oct	15 Oct	16 Oct	17 Oct	18 Oct	19 Oct	Total
Rows	639	600	639	600	635	600	600	**4,313**

Table 2: Final data-quality indicators

Indicator	Result	Indicator	Result
Unique products	799	Categories	3
Observation dates	7	Product-date rows	4,313
Duplicate product-date rows	0	Missing old prices	199
Missing ratings	25	Products seen all 7 days	469
Foreign-key violations	0	Database integrity	ok

The daily totals vary between 600 and 639 because catalogue composition and pagination changed. These differences are expected for a live marketplace and were not filled by copying observations forward. The 469 products present every day form the strongest panel for studying stability, while occasional listings still contribute to the overall market description.

7 Exploratory analysis approach

The analysis used three complementary units. First, product-date rows described daily listings. Second, unique-product summaries reduced repeated observations when comparing brands and countries. Third, first-to-last and consecutive-day comparisons measured changes only where both observations existed. This distinction prevents frequently visible products from dominating every result.

Outliers were retained because they may be genuine premium products. Price charts used a logarithmic scale where necessary. Correlations were calculated only for non-missing pairs, and their strength was interpreted cautiously. A displayed discount was also compared with the discount calculated from current and old prices, allowing small rounding differences.

8 Dataset limitations at analysis stage

Ratings and review counts are marketplace indicators, not controlled measurements of quality. Products can disappear because of sampling position, stock, seller action or ranking changes. Brand names were inferred from listing text and therefore include a large *Generic* class. These issues shape interpretation but do not invalidate the observed catalogue evidence.

9 Results: Jumia Nigeria

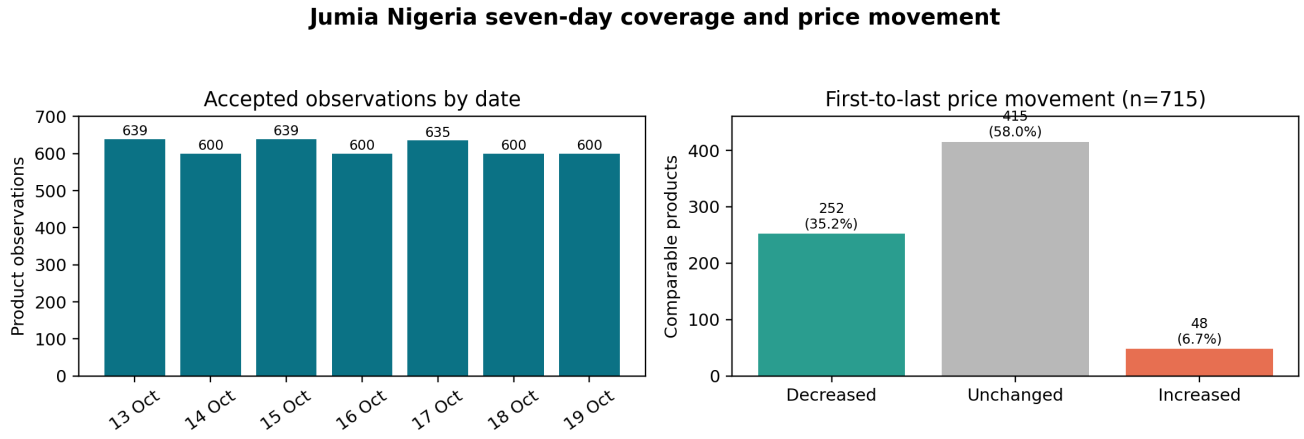


Figure 1: Coverage remained close to the target each day. Among comparable first and last observations, decreases were much more frequent than increases.

9.1 Price distributions and outliers

Prices were right-skewed within categories: most products occupied lower and middle price bands, while a smaller premium group produced large upper outliers. Consequently, category medians better represented a typical listing than means. Outliers were checked rather than automatically removed because premium phones, computers and electronics are plausible catalogue products.

9.2 Price, rating and popularity

The Spearman correlation between price and rating was $\rho = 0.110$, a very weak positive relationship. Review count and rating had $\rho = -0.215$, a weak negative relationship. Higher price therefore did not reliably imply a better rating, and heavily reviewed products were not necessarily rated higher. Review count is better treated as a visibility or engagement proxy than direct sales evidence.

9.3 Seven-day movement

Of 715 comparable products, 48 increased, 252 decreased and 415 were unchanged from first to last observation. Across all consecutive observations, 314 products changed price at least once, producing 378 recorded changes. For displayed discounts, 227 of 682 comparable products increased, 44 decreased and 411 remained unchanged. This suggests that promotional labels moved more often upward than downward during the study window, but the results do not reveal seller intent.

9.4 Brands, discounting and visibility

Generic had the largest unique-product count in every category: 127 in Computing, 103 in Electronics and 55 in Phones & Tablets. This dominance reflects marketplace labelling and many unbranded accessories. The additional group question examined visibility and stability: 469 products appeared on all seven days. Repeated visibility makes their movements more dependable, but visibility itself should not be interpreted as sales popularity without transaction data.

10 Cross-group comparison: Kenya and Nigeria

The comparison package supplied by Group 1 and Group 2 contained 553 unique-product summaries: 271 from Kenya and 282 from Nigeria. To improve comparability, the analysis focused on *Phones & tablets* and *Accessories* and converted both datasets to USD. The comparison period represented in the supplied file was 13–18 October, so it is a focused supplement rather than a replacement for the full Nigerian seven-day analysis.

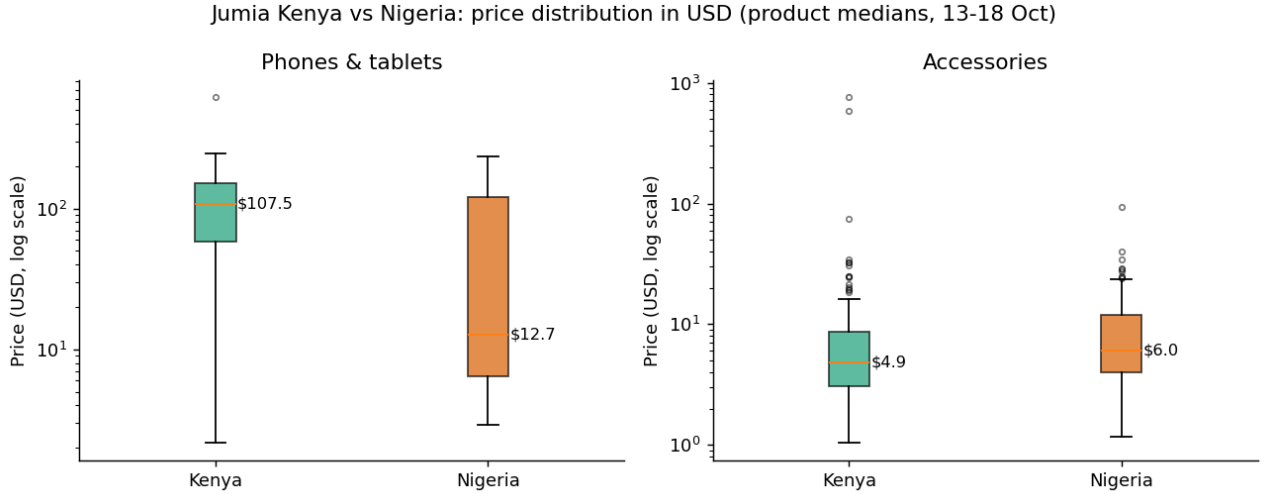


Figure 2: USD product-median distributions in the supplied comparison file. Log scaling reveals the wide spread and outliers.

The median phone/tablet price was $107.50 in Kenya and $12.67 in Nigeria. This large overall difference mainly reflects different product mixes: the Nigerian subset includes more low-cost phones and accessory-like entries within the broad class. Accessories were closer, with medians of $4.86 in Kenya and $6.00 in Nigeria. Overall country medians should therefore not be read as like-for-like price differences.

11 Cross-group comparison continued

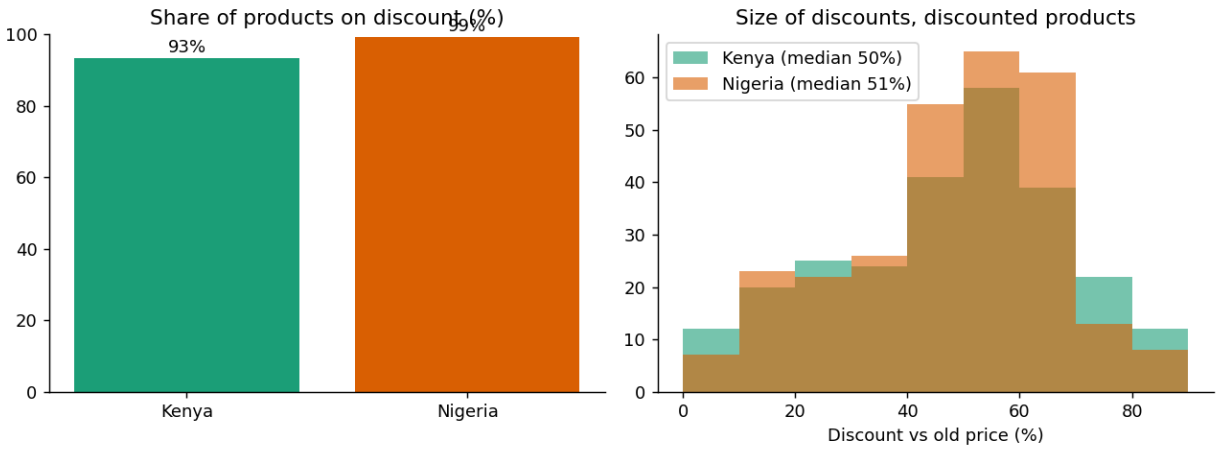


Figure 3: Discount prevalence and discount size for the comparable product-level subsets.

Discounting was very common in both samples: 93.4% of Kenyan summaries and 99.3% of Nigerian summaries were marked discounted. Median calculated discount among discounted items was approximately 50.0% in Kenya and 50.9% in Nigeria. These near-equal medians contrast with the difference in the product-mix price distributions.

The strongest cross-country evidence came from 38 closely matched listings. Twenty were cheaper in Nigeria and 18 were more expensive; the median Nigeria-to-Kenya price ratio was 0.976. Thus, the typical matched listing was about 2.4% cheaper in Nigeria, while individual gaps ranged from roughly 44% cheaper to 47% more expensive. This variation can arise from seller choice, promotions, exchange-rate treatment, taxes, logistics and stock. With only 38 matches, the evidence is useful but not sufficient for a general claim that one country is always cheaper.

Table 3: Focused Kenya–Nigeria comparison summary

Measure	Kenya	Nigeria
Unique-product summaries	271	282
Phones/tablets	60	71
Accessories	211	211
Median phone/tablet price (USD)	107.50	12.67
Median accessory price (USD)	4.86	6.00
Discounted share	93.4%	99.3%
Median discount, discounted items	50.0%	50.9%

12 Limitations and ethics

Limitations. The study covers seven days and selected catalogue pages, not the full Jumia marketplace. Listings can move between pages or disappear; seller, shipping and stock information may differ; brand extraction is imperfect; ratings are self-selected customer feedback; and displayed old prices may be promotional anchors. Currency conversion introduces dependence on the applied rate and timing. The Kenya comparison covers a narrower category set and six-day overlap, and only 38 listings were matched closely enough for like-for-like comparison. Findings are therefore descriptive snapshots rather than causal or nationwide retail conclusions.

Ethics. Only public product information was collected. The workflow checked `robots.txt`, identified itself, throttled requests and stopped on denial signals. It did not collect personal data, evade protection, authenticate, or fabricate missing observations. Raw files and checksums preserve an audit trail, while the report avoids claiming that review counts equal purchases or that short-term patterns predict future prices.

13 Conclusion

The project demonstrates a complete introductory data-mining workflow: reconnaissance, controlled web collection, validation, cleaning, normalized storage, exploratory analysis and communication. The seven-day Nigerian dataset is internally consistent and shows that stable prices were most common, price decreases clearly outnumbered increases, and ratings had little association with price. The cross-group exercise also demonstrates why product matching matters: unmatched category medians differed greatly, whereas matched Kenya–Nigeria listings were typically much closer in price. Future work could extend the collection period and improve product matching, but the present workflow appropriately answers the assignment without overstating the evidence.

14 Contribution summary

Member	Identifier	Attributed contribution
Damba Alex	S25B38/037	Project coordination, seven-day integration, final USD analysis, notebook and report preparation.
Mukisa Eric Williams	S25B38/025	Reconnaissance review, scraper/parser review and data-dictionary support.
Lwanga Darren Jordan	S25B38/022	Testing, validation checks and repository-quality review.
Bwambale Bisogho Sedrack	S25B38/038	Daily collection support, README updates and workflow documentation.
Masaba Carl Michael	S25B38/016	Daily collection support, quality reporting and database verification.

References and project evidence. DSC2107 Group Assignment Brief (2026); Jumia Nigeria public catalogue and `robots.txt`; Group 2 reconnaissance note, raw manifests, validation files, SQLite database and executed analysis notebook; Group 1/Group 2 Kenya–Nigeria comparison package supplied for this report.